Signals and Systems
Project Based Learning

Project Title:

EMG-Based Assistive Device for Enhanced Human Computer Interaction

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Problem Statements:

- There is a growing need for assistive technology that enables individuals with limited mobility to interact with computers more effectively.
- Existing Human-Computer Interaction (HCI) systems often rely on fine motor skills, which may not be suitable for users with muscular impairments.
- People with disabilities like amputated wrist or arms have a difficult time interacting with electronic devices which primarily use touch or mouse input.
- They lack the motor function to properly use the devices to the full extent

Solution:

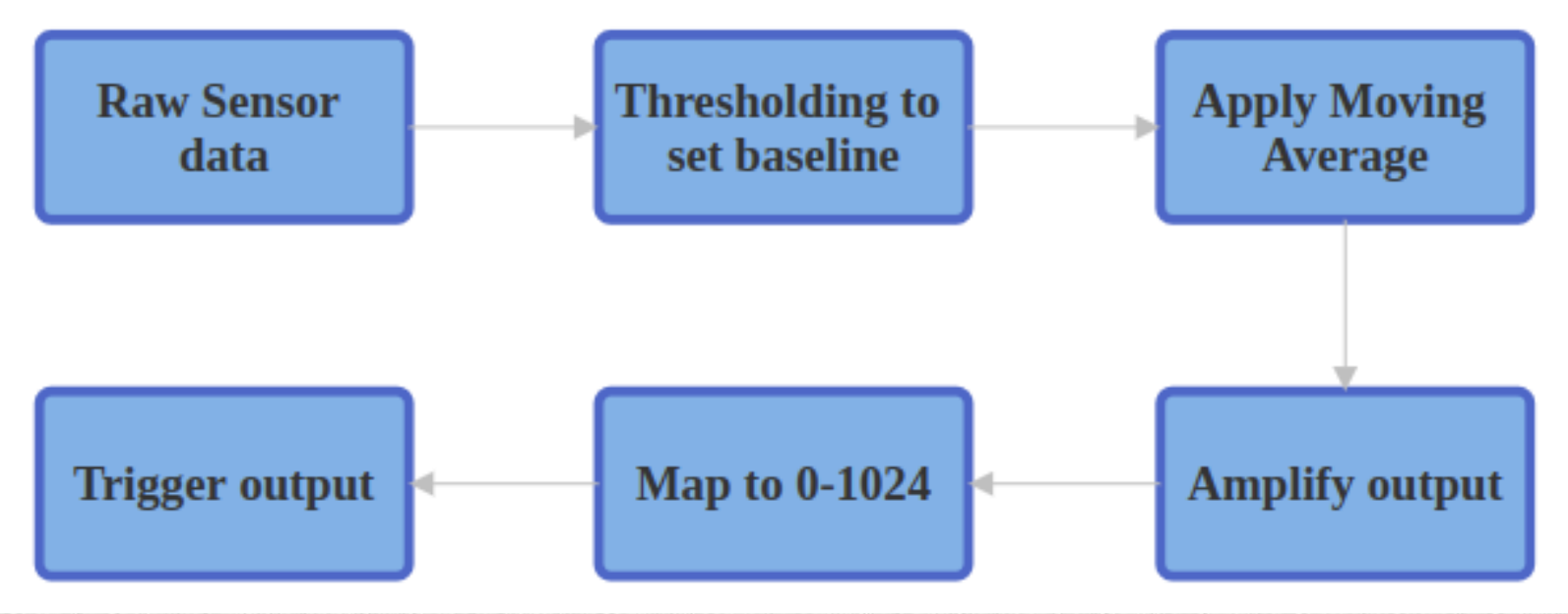
- Make a interface which can use muscles from different parts of the body as an input to an electronic device

Objectives

- To develop a system that uses EMG signals to control a computer mouse.
- To implement techniques for signal processing, including noise filtering and thresholding, to ensure accuracy of the input signal
- To test and validate the system’s usability in real-world scenarios.

Process

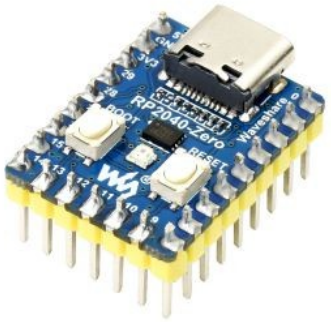
- Acquire EMG Signal.
- Perform signal processing, Bandpass, Filtration, Amplification, etc.
- Send data to computer and process mouse movement.



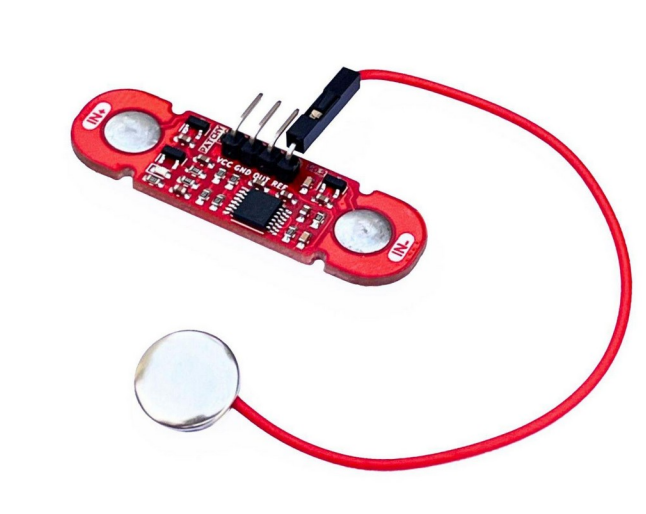
Implementation: Hardware

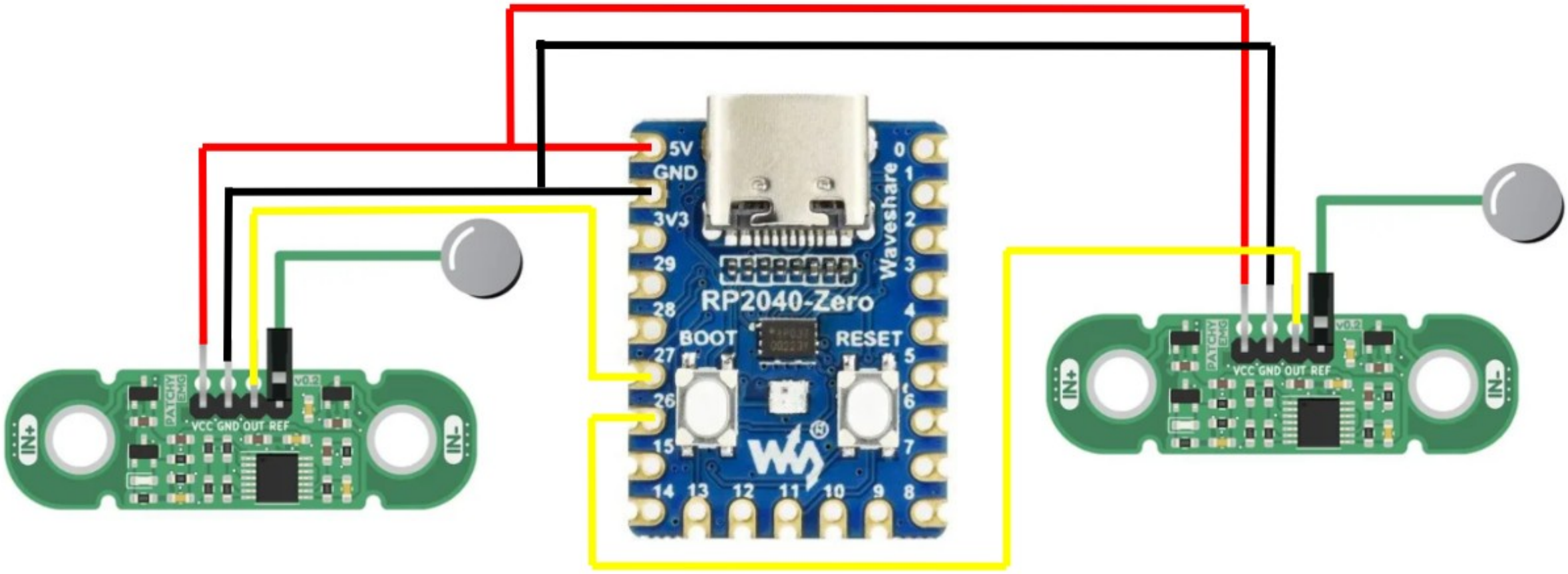
Components:

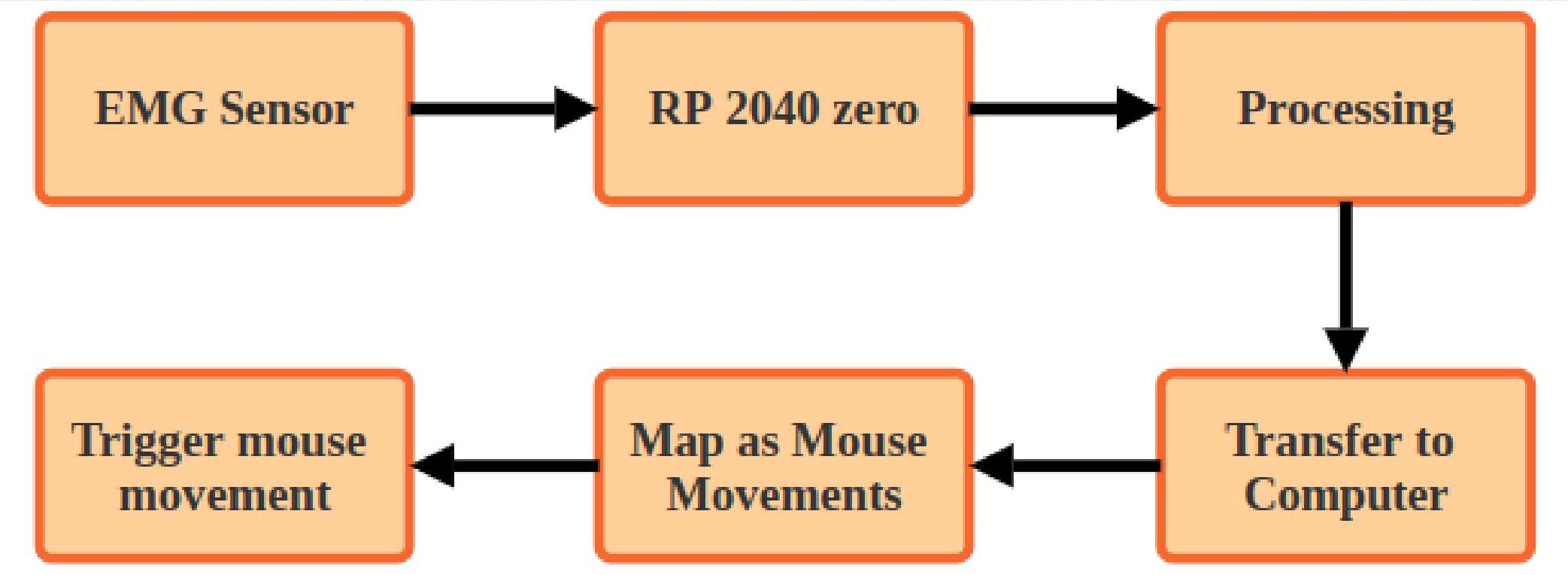
Waveshare RP2040 zero- Low cost, low power, small form factor micro controller with powerful Arm Cortex M0+ cores at 133Mhz and 12 bit ADC for higher resolution analog signal processing



Upside down labs Muscle Bioamp Patchy- 2 wearable EMG sensors with an integrated band pass filter 72-720Hz




5V
GND
3V3
29
28
27
26
15
14 13 12 11 10 9 8
0 1 2 3 4 5 6 7
Waveshare
RP2040-Zero
BOOT
RESET
VCC GND OUT REF
IN+
IN-


EMG Sensor
RP 2040 zero
Processing
Transfer to Computer
Map as Mouse Movements
Trigger mouse movement

Implementation: Software

Arduino IDE- Simple and easy to use, wide support for multiple libraries for microcontrollers



Python 1.12-

Powerful programming language with libraries for extended control on microcontrollers and computers

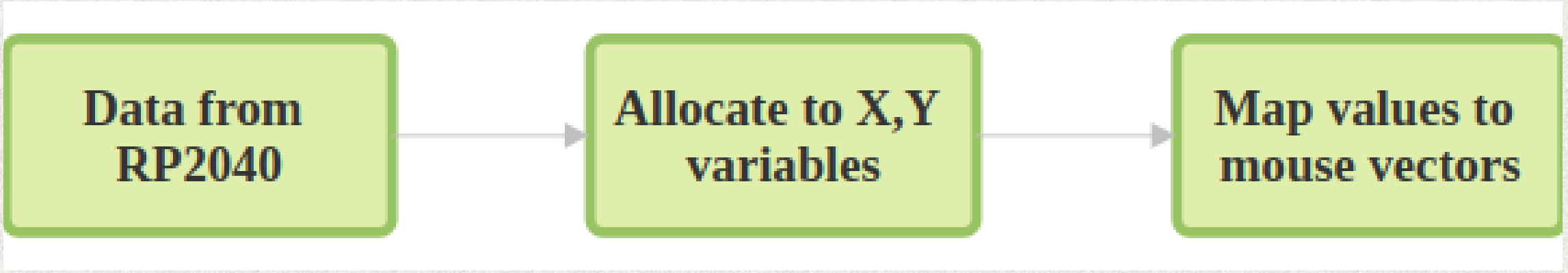


Libraries used:

serial

pyautogui

time, csv


Data from RP2040
Allocate to X,Y variables
Map values to mouse vectors

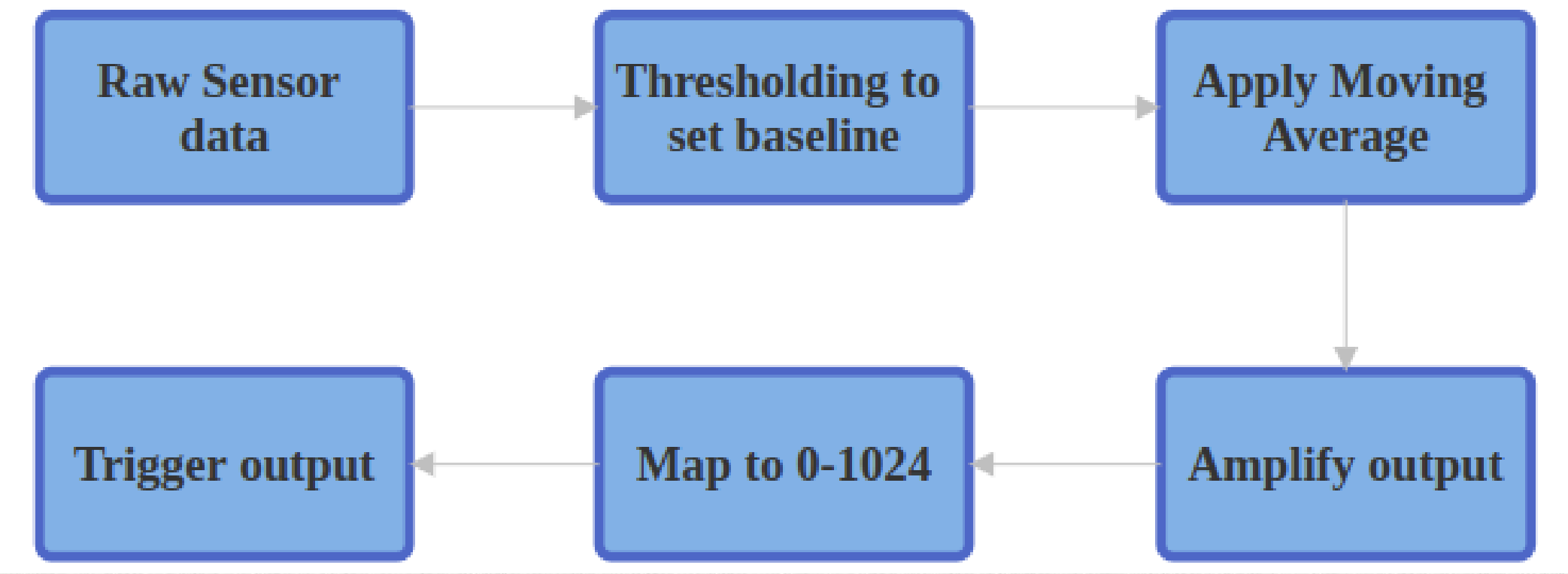
Implementation: Signal Processing

Thresholding: Only strong signals (indicating deliberate muscle activity) were passed to control the mouse.

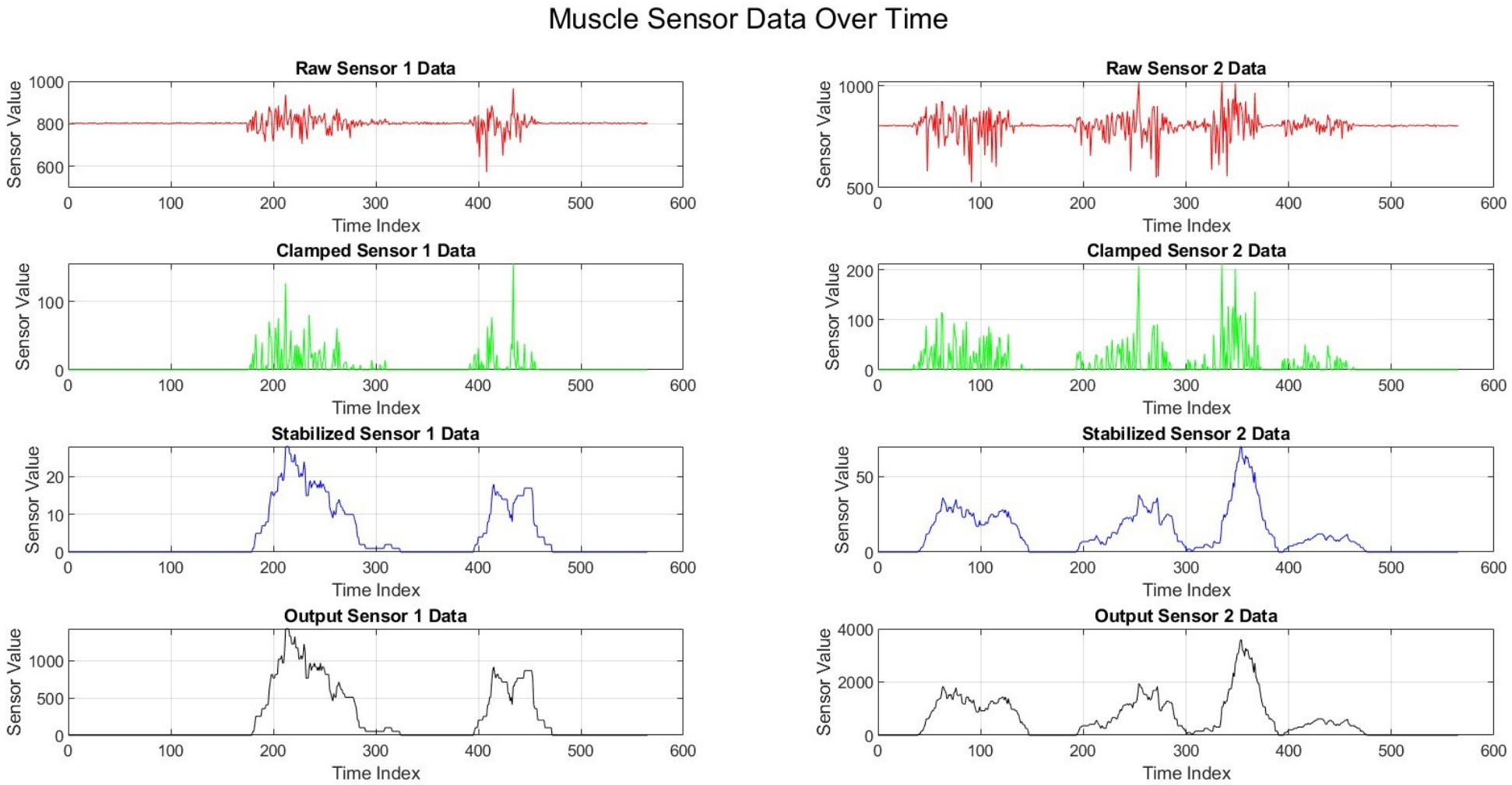
Noise Filtering and smoothing: **Moving average filter** was applied to smooth out the EMG signal and reduce artifacts.

Signal Amplification: As EMG sensor signals are weak(~3.3v), they were **amplified** before processing to make sure muscle movements were detected accurately.

Real-time Processing: Signal processing was done in real-time to ensure immediate feedback and smooth mouse control.


Raw Sensor data
Thresholding to set baseline
Apply Moving Average
Amplify output
Map to 0-1024
Trigger output

Implementation: Signal Processing



Testing & Validation:

- EMG signal processing was verified by using the **serial plotter** available in **Arduino IDE**
- Serial communication was tested using python **serial library** and data was saved to a csv file and hence verified
- EMG signals were sent through serial communication and verified by plotting rhe **resulting CSV file** in **MATLAB**
- mouse movement was tested by using a python script with the **pyautogui library**

Final Report & Analysis

- **Key Takeaways**
- EMG signals can be used to provide computer inputs without using traditional methods
- signals accuracy and responsiveness are good but still can be improved
- They can also be used for applications other than the computer control like robotics mechanical arm control etc

Future Upgrades/Possibilities

- More sensors for multi channel control, making more advanced control available
- Simplifying the program and algorithms for efficient and faster execution
- Getting better signals by testing different filters like boxcar and gaussian smoothing
- Adding on the go adjustable sensitivity for user’s ease of use
- Implementing a mouse mapping algorithm for control without extreme muscle contraction.

Conclusion

- Successfully Acquired EMG sensor data and applied signal processing like amplification and Filtering.
- Achieved EMG to mouse movement by mappint the processed output to mouse vector
- Found potential for improvement and learnt about different algorithms and methods for signal processing

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Thank You